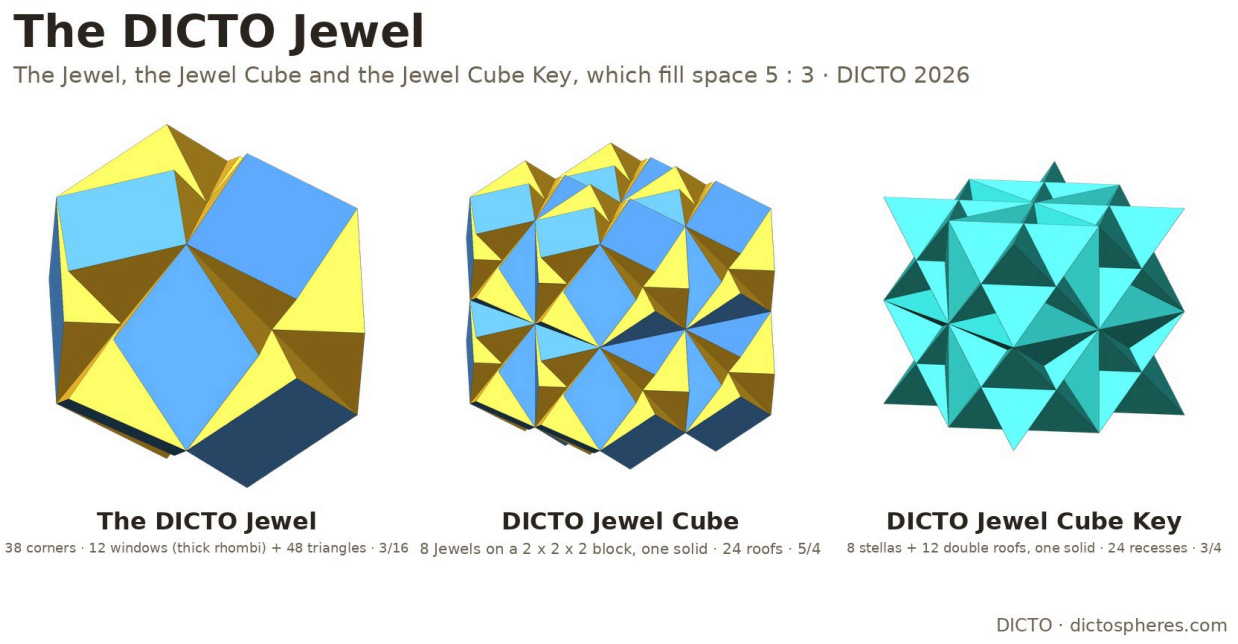
The DICTO Jewel

Euclid's dodecahedron carved by its neighbours' stella octangulas: an exact solid with twelve Penrose windows, its lattice, its clusters and rhombohedron, and the DICTO Jewel Cube and Jewel Cube Key, which fill space 5 : 3

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Version 1 · 20 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/3595a60/studies/2026-10-20-dicto-jewel (research branch, commit 3595a60; also attached to this record) · Interactive: dictospheres.com → Kaleidohedra → DICTO → Stella–Jewel Lattice

The Jewel, its stella-octangula counterpart, its name, its clusters, the Jewel Cube and the name Jewel Cube Key are DICTO's (findings #10, #10b and #15 on dictospheres.com); computation, exact proofs and drafting with Claude (Anthropic), which also found the Jewel Cube Key as the space between Jewel Cubes.



Abstract. Put Euclid's dodecahedron on a cube, as roofs on its six faces, and carve away the stella octangulas of the six neighbouring cubes: what remains is the **DICTO Jewel**, with 38 corners and 60 faces, twelve of them thick Penrose rhombi (the windows) and 48 triangles, and volume exactly 3/16 of the Jewel Cube defined below (or 12 on cells of edge 2). With stella octangulas on the other cells, Jewels fill space as a rock-salt pattern, 3 : 1 by volume [2]; we prove this exactly: each stella's cube is the stella and six carved roofs of exactly 1/96 each. The Jewels form two clusters, a tetrahedral one (4 Jewels, 228 faces) whose copies share out all Jewels, and an octahedral one (6 Jewels round a sealed stella hole) that tiles space with loose stellas on exactly 12 lattices of index 12; eight Jewels on the corners of the face-centred cubic cell, meeting on 18 windows, form a rhombohedron round one stella. Eight Jewels on a 2 × 2 × 2 block of cells, overlapping inside, make one solid, the **DICTO Jewel Cube** (5/4: a cube with 24 roofs); the space between Jewel Cubes is another single solid, the **DICTO Jewel Cube Key** (3/4: eight stellas and twelve double roofs), whose 24 recesses hold the Cubes' 24 roofs. Cubes and Keys fill space 5 : 3, the rock-salt pattern of the Jewel one level up, and every Cube meets each of its 12

> neighbours on exactly two windows. As a solid the Jewel is pyritohedral, so it twins like the Ogdoecontahedron [5]; its packing cannot, because a (111) mirror does not keep the cube lattice. All claims are exact in $\mathbb{Q}(\sqrt{5})$ except where marked float.

1. The Jewel (exact)

Frame. As in the EKP cell [1] and [2]: cells are cubes centred on a cubic lattice; a cell is *even* or *odd* by the parity of its index sum, and the even cells form a face-centred cubic lattice. All values below are at the scale where the Jewel Cube (§4) has edge 1, so a cell has edge 1/2 and volume 1/8; the scripts use cells of edge 2, where every volume is 64 times larger.

Construction. Euclid builds the regular dodecahedron on a cube by setting a roof on each face [3]: the hull of the face and a ridge of length $1/\varphi$ times the cell edge, at height $(\varphi - 1)/2$ times the cell edge; each roof is $(1 + \sqrt{5})/192$ and the dodecahedron $(5 + \sqrt{5})/32$. Each roof reaches into the next cell, where the stella octangula (Kepler's two tetrahedra on the cube's corners [4], volume 1/16) takes away $(\sqrt{5} - 1)/192$ of it, leaving a **carved roof** of exactly 1/96. The Jewel is its cell and its six carved roofs: $1/8 + 6/96 =$ **3/16**.

Its surface. 38 corners: the cube's 8, the 12 ridge points, the 6 face centres, and 12 points that complete the windows. Its 60 faces are 12 **windows**, one across each cube edge, lying in the dodecahedron's face planes, and 48 triangles in the stellas' planes; $38 - 96 + 60 = 2$. Each window is a thick Penrose rhombus: side $(\sqrt{5} - 1)/4 = 1/(2\varphi)$, long diagonal 1/2 (the cell's edge), angle 108° at the ridge corners ($\cos 108° = (1 - \sqrt{5})/4$, exact) and 72° at the cube corners [6]. The Jewel's corners are centrosymmetric, have the three-fold turns and the three coordinate mirrors, and no mirror across a (111) plane: it is pyritohedral.

2. The Stella–Jewel Lattice (exact)

Jewels on the even cells and stella octangulas on the odd cells fill space, $3/16 : 1/16 = 3 : 1$ [2]. The first check (finding #10b) sampled points; here it is exact. Take an odd cell. The six Jewels on its face-neighbours each reach into it with one carved roof (their ridges are Jewel corners). These six roofs lie in the cell, any two meet only in a face (the intersection of the uncarved roofs has volume exactly 0), and each carved part is 1/96. With the stella they give $1/16 + 6/96 = 1/8$, the cell: so an odd cell is exactly its stella and six carved roofs, an even cell is the core of its Jewel, and the Jewels and stellas tile space.

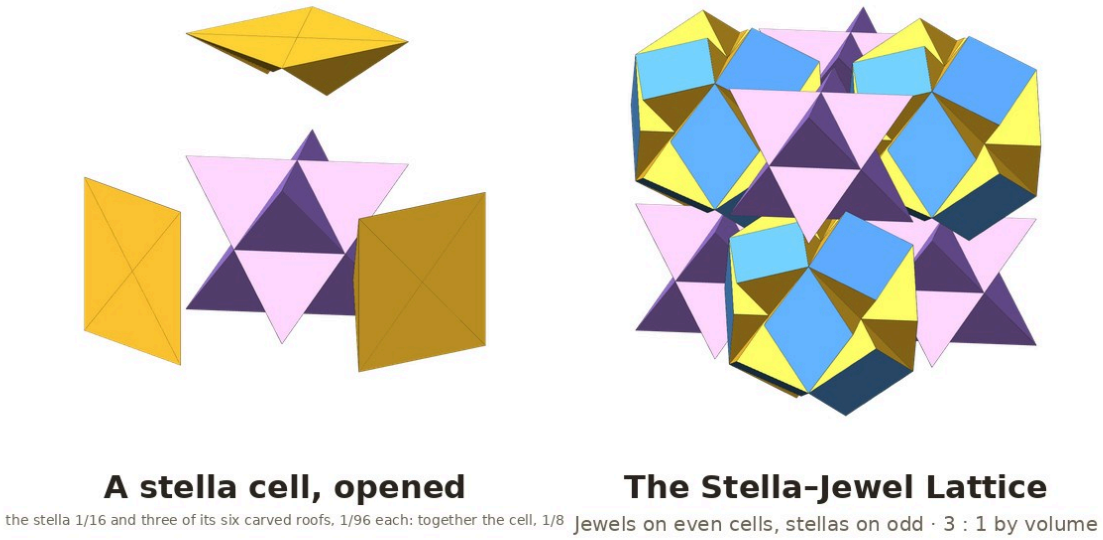


Figure 1. Left: an odd cell opened, its stella octangula (violet) and three of the six carved roofs that the neighbouring Jewels push into it. Right: a 2 × 2 × 2 block of the Stella–Jewel Lattice, drawn slightly apart.

3. Clusters and the rhombohedron (exact)

Neighbours. Two Jewels on neighbouring even cells (one step along a face diagonal) share exactly one window; Jewels two cells apart along an axis share none.

The tetrahedral cluster (finding #15): four Jewels on mutually neighbouring even cells. Each pair shares one window, so its surface has $4 \times 60 - 2 \times 6 =$ **228 faces** (192 triangles, 36 windows), and all four meet at one point, their cells' common corner; volume 3/4. Anchored on the cells whose three indices are all even, these clusters use every Jewel exactly once.

The octahedral cluster (finding #15): the six Jewels round an odd cell. Each shares a window with its four neighbours in the ring (12 windows), so its outer surface has **288 faces** (240 triangles, 48 windows), and inside it the 48 walls of the Jewels' carved roofs enclose the stella-shaped hole, sealed but for its eight tips; volume 9/8, or 19/16 with the hole filled. Octahedral clusters tile space together with loose stellas on exactly **12 lattices of index 12** (exact integer search over all lattices of that index), for example the one with Hermite basis (1,0,3), (0,1,5), (0,0,12) in cell units; each period holds one cluster and five loose stellas.

The rhombohedron [2]: eight Jewels on the corners of the face-centred cubic cell. Its faces have 60° angles, so the Jewels meet not only along its 12 edges but also along the 6 short face diagonals: **18 windows**. All six face-neighbours of the odd cell at its centre are among its corners, so that cell's stella is sealed inside: 8 Jewels and a stella, 25/16 = 8⅛ Jewels.



Figure 2. The tetrahedral cluster, the octahedral cluster and the Jewel rhombohedron (windows blue).

4. The DICTO Jewel Cube and the DICTO Jewel Cube Key (exact)

The Jewel Cube. Put a whole Jewel on each of the eight cells of a 2 × 2 × 2 block, on the even and the odd cells alike. Inside the block they overlap; outside it, each of the 24 outer cell faces carries its Jewel's roof. As one solid, the **DICTO Jewel Cube** is the block (edge 1) with 24 carved roofs on its faces: 1 + 24/96 = **5/4**, six and two-thirds Jewels.

The Jewel Cube Key. Put Jewel Cubes on alternate blocks, rock-salt fashion (the blocks of even index sum). Each other block is then entered by 24 roofs, four through each face, from its six neighbouring Cubes; what is left is one solid, the **DICTO Jewel Cube Key**: 1 − 24/96 = **3/4**, the volume of four Jewels. It is eight stella octangulas, one per cell, joined by twelve double roofs across the inner cell faces: 8/16 + 12 × 2/96 = 3/4. Its 24 recesses are exactly the Cubes' 24 roofs, so Cubes and Keys fill space, **5 : 3**.

The same pattern one level up. A cell with six carved roofs added is a Jewel (1/8 + 6/96), with six taken away a stella (1/8 − 6/96); a block with 24 carved roofs added is a Jewel Cube, with 24 taken away a Jewel Cube Key; and both levels sit in the rock-salt pattern [7].

Windows. The Cubes sit on a face-centred cubic lattice of blocks. Each Cube meets each of its 12 neighbours on exactly **two whole windows** (24 in all, exact), and no two Cubes' roofs overlap. The Jewel Cube is a union of overlapping Jewels, which is why it is defined as one solid.

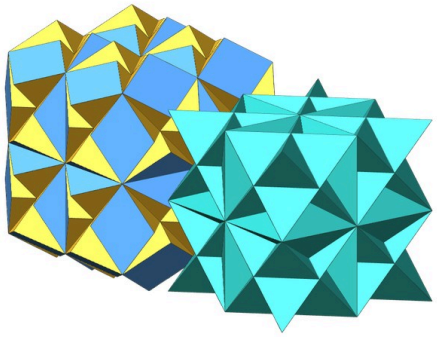


Figure 3. A Jewel Cube and the Jewel Cube Key next to it in the Jewel Cube Lattice, drawn out a little: the Cube's roofs go into the Key's recesses.

5. Twins

By the twin rule [5], a centrosymmetric solid with a three-fold axis has as its mirror image in a plane at right angles to that axis the same solid turned 60°. The Jewel is such a solid, so as a solid it twins like the DICTO Ogdoecontahedron. Its packing does not: in [8] a twin through a (111) layer of the Stella–Jewel Lattice left about 4% of sampled points uncovered and 4% covered twice (float). The reason is exact and simple: the mirror in a (111) plane takes the cubic lattice of cells to a different lattice (the step to the next cell, (1, 0, 0) in cell units, goes to (1/3, −2/3, −2/3)), so the mirrored cells cannot meet the cells on the other side face to face. The Ogdoecontahedron's packing twins because its cells, octahedra, are mirrored into the next octahedra of the gyrated octet; the Jewel's cells are cubes, and a mirrored cube does not continue the cubic lattice.

6. Prior art and what is claimed

- **Classical parts.** The dodecahedron as a cube with six roofs is Euclid's (Elements XIII.17) [3]; the stella octangula is Kepler's [4]; the thick rhombus is Penrose's [6]; the rock-salt arrangement is that of sodium chloride [7]; the EKP cell holds them together [1].
- **Lalvani.** US 4,723,382 [9] builds periodic and non-periodic structures from polygonal members whose edges all lie on the fifteen two-fold axes of the icosahedral group. All 48 window edges of the Jewel lie on those axes, but none of its other 48 edges do (lalvani_edges.py, float), so the Jewel, and with it the stella and the lattices here, fall outside that edge condition (our reading, not legal advice).
- **Already published.** The Jewel's definition, its volume and the Stella–Jewel Lattice are in [2], with the cluster networks; the Jewel and the lattice were first released with the Kaleidohedra app (findings #10, #10b, #15 [10]).

- **Claimed here:** the exact proofs of the Jewel's surface, volume and symmetry and of the Stella–Jewel tiling; the clusters as exact solids, with the 12 lattices of octahedral clusters; the 18 windows of the rhombohedron; the DICTO Jewel Cube and the DICTO Jewel Cube Key, their 5 : 3 tiling, the two-window contacts and the pattern one level up; and the exact reason the packing does not twin.

In the sources checked (the full text of the 80 library sources listed in the appendix, and web searches on 20 October 2026) we did not find the Jewel, its clusters, the Jewel Cube or the Jewel Cube Key.

7. Further work

- The Jewel Cube and the Jewel Cube Key in the Kaleidohedra app, beside the Stella–Jewel Lattice.
- A third level: Jewel Cubes on a 2 × 2 × 2 block of blocks.
- A physical puzzle: Cubes and Keys interlock and can be cut in wood or printed.

Data and reproducibility

Exact ($\mathbb{Q}(\sqrt{5})$): jewel_exact.py (roof, Jewel, Stella–Jewel tiling, Jewel Cube and Key, windows between Cubes, the (111) mirror; log jewel-exact.log, 18 PASS) and cluster_exact.py (clusters, the 12 lattices, the rhombohedron; log cluster-exact.log, 11 PASS), with the exact tools of [11]. Float: lalvani_edges.py. Figures: jewel_figs.py (float copies of the exact pieces), shots.mjs, jewel_compose.py.

References

1. DICTO, The Euclid–Kepler–Pacioli Cell, DICTO DOI: 10.5281/zenodo.23284106.
2. DICTO, Kagome Hulls, DICTO DOI: 10.5281/zenodo.23376317.
3. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III (Books X–XIII), Cambridge (1908), Book XIII, Proposition 17.
4. J. Kepler, Harmonices mundi libri V (1619), Book II.
5. DICTO, The DICTO OG Family on Every Stacking, DICTO DOI: 10.5281/zenodo.23490114.
6. R. Penrose, Pentaplexity: a class of non-periodic tilings of the plane, Mathematical Intelligencer 2, 32–37 (1979).
7. W. L. Bragg, The structure of some crystals as indicated by their diffraction of X-rays, Proc. R. Soc. Lond. A 89, 248–277 (1913).
8. DICTO, The DICTO Ogdoecontahedron on the Gyrated Octet, DICTO DOI: 10.5281/zenodo.23484012.
9. H. Lalvani, Building structures based on polygonal members and icosahedral symmetry, US Patent 4,723,382 (1988).
10. DICTO, Kaleidohedra by DICTO, DICTO DOI: 10.5281/zenodo.23173809; releases 10.5281/zenodo.23223555 (#10b) and 10.5281/zenodo.23257393 (#15, #16).
11. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016 (the exact tools q5.py and exactpoly.py).

Appendix. The reference library searched

The full text of the 80 sources below in DICTO's reference library was searched on 20 October 2026 for: Penrose, stella octangula, rock salt, Lalvani, Euclid, roof and rhombic. None contains the results of this paper. The relevant hits are Euclid and Kepler [3, 4], Pacioli, Messer's stellations of the rhombic triacontahedron (the stella octangula), Lalvani's NASA report on octahedral–tetrahedral trusses, and the papers on Penrose tilings (Kramer, Baake and Grimm, Dietl and Eschenburg). Thirteen further items are page images without a text

layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, among them US 4,723,382, read separately in full on 8 October 2026, and Fedorov 1885).

Web searches (20 October 2026): a dodecahedron carved by stella octangulas; space-filling pairs of a dodecahedron-based solid and the stella octangula; cubes of overlapping dodecahedra.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings

S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)